BRIDGE AT CHAINAGE (PADLI GUJARESHWAR)

**ABSTRACT OF COST**

| S.No | Particulars. | Quantity | | Rate | Unit | Amount |
|---|---|---|---|---|---|---|
| 1 | 1 Earthwork in excavation for foundation of B-1 structure complete with all lift, dewatering, 1.1.4 shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soll within a lead of 1000 meters, as per drawing and technical Earth In excavation for foundation of specification. [MORTH Specification: Clause 304]. in ordinary soll [MORTH Specification clause 301] BY MECHANICAL MEANS Depth upto 3m | 618.58 | Cum | 57 | P Cum | 35259.00 |
| 2 | Earthwork In excavation for foundation of B-1 structure complete with all lift, dewatering, 1.3.2 shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth in excavation for foundation of specification. [MORTH Specification Clause 304].In Ordinary/Soft rock (not requiring blasting) [MORTH Specification: Clause 3011 BY MECHANICAL MEANS | 572.21 | Cum | 83 | P Cum | 47493.02 |
| 3 | Earthwork in excavation for foundation of B-1 structure complete with all lift, dewatering. 1.4.2 shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth in excavation for foundation of specification. [MORTH Specification: Clause 304]. In Hard Rock [MORTH Specification : Clause 301, 302, 303]. Blasting Prohibited | 1088.31 | Cum | 670 | P Cum | 729167.70 |
| 4 | Structural cement concrete (Design Mix) in open foundations, with form work, at level. Cement concrete for plain/reinforced concrete in open foundations as per drawing and technical specification. [MORTH Specification: Section 1700 & 1500).<br>With 20mm nominal size aggregate<br>P.C.C. M-20 grade | 114.44 | Cum | 4017 | P Cum | 459709.50 |
| 5 | Structural cement concrete (Design Mix)<br>B-2 in open foundations, with form work, at 6.6.6 any level. Cement plain/reinforced concrete for concrete in open foundations as per drawing and technical specification. [MORTH Specification : Section 1700 & 1500].<br>With 20mm nominal size aggregate<br>R.C.C. M-25 grade | 961.61 | Cum | 4341 | P Cum | 4174341.41 |

| S.No | Particulars. | Quantity | | Rate | Unit | Amount |
|---|---|---|---|---|---|---|
| 6 | Providing and laying steel reinforcement at any level in foundation complete as per 13.1 drawing and technical specification. [MORTH Specification: Section 1600]. HYSD bar reinforcement | 64.55 | T | 88567 | P tonne | 5716734.15 |
| 7 | Structural cement concrete (Design Mix) in open foundations, with form work, at level. Cement concrete for plain/reinforced concrete in open foundations as per drawing and technical specification. [MORTH Specification: Section 1700 & 1500).<br>With 20mm nominal size aggregate<br>P.C.C. M-15 grade | 272.80 | Cum | 3038 | P Cum | 828766.40 |
| 8 | Structural cement concrete for plain concrete/ reinforced concrete for substructure at any level, complete as per [MORTH Specification: Section 1500 & 1700]. drawing and technical specification. Design mix concrete PCC M-25 Upto 5 mtr. Height | 372.09 | cum | 4319 | P Cum | 1607050.23 |
| 9 | Steel reinforcement in substructure<br>HYSD reinforcement | 57.16 | T | 88737 | P tonne | 5072118.18 |
| 10 | Providing A.C. pipe Weep holes 150mm B-3 dia in Masonry/plain Concrete/Reinforced 13 Concrete abutment, wing wall/return wall complete as per drawing and technical specification. [MORTH Specification : Clause 2205]. | 408.50 | R mtr | 114 | P RM | 46569.00 |
| 11 | Providing and and filling in foundation Btrenches, and back filling behind abutments, wing walls, return wall etc., and below pipe bed in layers not exceeding 150mm thick including watering and compacting (IRC-78) | 10.11 | cum | 745 | P Cum | 7534.19 |
| 12 | Filter medium behind abutment, wing wall<br>and return wall, complete as per drawing<br>and technical specification. (Same as R7.20.3)<br>[MORTH Specification: Clause 2504]. | 419.13 | cum | 560 | P Cum | 234712.80 |
| 13 | Structural RCC/PSC upto a height of 5 mtr. for reinforced concrete in superstructure complete including steel formwork, scaffolding etc. As per drawing and technical specification.Design mix concrete. [MORTH Specification : 1500 & 1700].RCC M-30 grade Solid slab superstructure at any level in superstructure complete as per drawing and technical Specification. [MORTH Specification: Section 1600] | 297.09 | cum | 6307 | P Cum | 1873752.94 |
| 14 | Steel reinforcement in Superstructure<br>HYSD reinforcement | 27.06 | T | 89654 | P tonne | 2426395.86 |

| S.No | Particulars. | Quantity | | Rate | Unit | Amount |
|---|---|---|---|---|---|---|
| 15 | Reinforced cement concrete wearing coat B-4 M-30 grade at any level including formwork and reinforcement complete as per drawing and technical Specification. MORTH Specification: Clause 2204, 1500 & 1700]. | 43.56 | Cum | 7720 | P Cum | 336302.50 |
| 16 | Providing and Laying PCC M-15 grade B-4 leveling course below approach slab 15 complete as per drawing and technical specifications. [MORTH specification: 2700] | 14.22 | cum | 2471 | P Cum | 35137.62 |
| 17 | Reinforced cement concrete M-30 grade B-4 approach slab including reinforcement 16 and formwork complete as per drawing and technical specifications. [MoRTH specification: 1500, 1600, 1700, 2704] | 21.33 | cum | 7073 | P Cum | 150867.09 |
| 18 | Providing, laying and fixing of Filler Joint Providing and fixing 2mm thick Corrugated Copper Plate in expansion joint complete as per drawing and technical specifications in MORTH specification: clause 2605. | 31.88 | Sqm | 3673 | P Sqm | 117076.88 |
| | | | | Total | Rs. | 18959455.58 |

**BRIDGE AT CHAINAGE (PADLI GUJARESHWAR)**

DETAILS OF MEASUREMENTS

| S.N. | Particulars | Nos. | | Length | | Breadth | | Height | | Qty. | Units |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Earthwork in excavation for foundation of structure complete with all lift, dewatering, 1.1.4 shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soll within a lead of 1000 meters, as per drawing and technical Earth In excavation for foundation of specification. [MORTH Specification: Clause 304]. in ordinary soll [MORTH Specification clause 301] BY MECHANICAL MEANS Depth upto 3m | | | | | | | | | | |
| | Abutment A1 | 1 | x | 8.50 | x | 8.80 | x | 1.50 | = | 112.20 | Cum |
| | Piers | 1 | x | 5.90 | x | 13.10 | x | 1.65 | = | 127.53 | Cum |
| | | 1 | x | 5.90 | x | 13.10 | x | 1.65 | = | 127.53 | Cum |
| | | 1 | x | 5.90 | x | 13.10 | x | 1.80 | = | 139.12 | Cum |
| | Abutment A2 | 1 | x | 8.50 | x | 8.80 | x | 1.50 | = | 112.20 | Cum |
| | | | | | | **Total Qty.** | | | | **618.58** | **Cum** |
| 2 | Earthwork In excavation for foundation of B-1 structure complete with all lift, dewatering, 1.3.2 shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth in excavation for foundation of specification. [MORTH Specification Clause 304].In Ordinary/Soft rock (not requiring blasting) [MORTH Specification: Clause 3011 BY MECHANICAL MEANS | | | | | | | | | | |
| | Abutment A1 | 1 | x | 8.50 | x | 8.80 | x | 1.50 | = | 112.20 | Cum |
| | Piers | 1 | x | 5.90 | x | 13.10 | x | 1.50 | = | 115.94 | Cum |
| | | 1 | x | 5.90 | x | 13.10 | x | 1.50 | = | 115.94 | Cum |
| | | 1 | x | 5.90 | x | 13.10 | x | 1.50 | = | 115.94 | Cum |
| | Abutment A2 | 1 | x | 8.50 | x | 8.80 | x | 1.50 | = | 112.20 | Cum |
| | | | | | | **Total Qty.** | | | | **572.21** | **Cum** |
| 3 | Earthwork in excavation for foundation of B-1 structure complete with all lift, dewatering. 1.4.2 shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth in excavation for foundation of specification. [MORTH Specification: Clause 304]. In Hard Rock [MORTH Specification : Clause 301, 302, 303]. Blasting Prohibited | | | | | | | | | | |
| | Abutment A1 | 1 | x | 8.50 | x | 8.80 | x | 2.25 | = | 168.30 | Cum |
| | Piers | 1 | x | 5.90 | x | 13.10 | x | 3.00 | = | 231.87 | Cum |

| S.N. | Particulars | Nos. | | Length | | Breadth | | Height | | Qty. | Units |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 | x | 5.90 | x | 13.10 | x | 3.15 | = | 243.46 | Cum |
| | | 1 | x | 5.90 | x | 13.10 | x | 2.85 | = | 220.28 | Cum |
| | Abutment A2 | 1 | x | 8.50 | x | 8.80 | x | 3.00 | = | 224.40 | Cum |
| | | | | | | **Total Qty.** | | | | **1088.31** | **Cum** |
| 4 | Structural cement concrete (Design Mix) in open foundations, with form work, at level. Cement concrete for plain/reinforced concrete in open foundations as per drawing and technical specification. [MORTH Specification: Section 1700 & 1500). With 20mm nominal size aggregate P.C.C. M-20 grade | | | | | | | | | | |
| | Abutment A1 | 1 | x | 8.50 | x | 8.80 | x | 0.30 | = | 22.44 | Cum |
| | Piers | 1 | x | 5.90 | x | 13.10 | x | 0.30 | = | 23.19 | Cum |
| | | 1 | x | 5.90 | x | 13.10 | x | 0.30 | = | 23.19 | Cum |
| | | 1 | x | 5.90 | x | 13.10 | x | 0.30 | = | 23.19 | Cum |
| | Abutment A2 | 1 | x | 8.50 | x | 8.80 | x | 0.30 | = | 22.44 | Cum |
| | | | | | | **Total Qty.** | | | | **114.44** | **Cum** |
| 5 | Structural cement concrete (Design Mix) B-2 in open foundations, with form work, at 6.6.6 any level. Cement plain/reinforced concrete for concrete in open foundations as per drawing and technical specification. [MORTH Specification : Section 1700 & 1500]. With 20mm nominal size aggregate R.C.C. M-25 grade | | | | | | | | | | |
| | Abutment A1 | 1 | x | 8.20 | x | 8.50 | x | 0.85 | = | 59.25 | Cum |
| | | 1 | x | 4.75 | x | 8.50 | x | 1.80 | = | 72.68 | Cum |
| | Piers | 1 | x | 5.60 | x | 12.80 | x | 3.10 | = | 222.21 | Cum |
| | | 1 | x | 5.60 | x | 12.80 | x | 3.25 | = | 232.96 | Cum |
| | | 1 | x | 5.60 | x | 12.80 | x | 3.30 | = | 236.54 | Cum |
| | Abutment A2 | 1 | x | 8.20 | x | 8.50 | x | 0.85 | = | 59.25 | Cum |
| | | 1 | x | 4.75 | x | 8.50 | x | 1.95 | = | 78.73 | Cum |
| | | | | | | **Total Qty.** | | | | **961.61** | **Cum** |
| 6 | Providing and laying steel reinforcement at any level in foundation complete as per 13.1 drawing and technical specification. [MORTH Specification: Section 1600]. HYSD bar reinforcement | | | | | | | | | | |
| | Abutment A1 & A2 | 2 | x | 11215 | | | | | = | 22430 | Kg |
| | Piers | 3 | x | 14039 | | | | | = | 42117 | Kg |
| | | | | | | **Total Qty.** | | | | **64547** | **Kg** |
| | | | | | | | | | | **64.55** | **MT** |

| S.N. | Particulars | Nos. | | Length | | Breadth | | Height | | Qty. | Units |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 7 | Structural cement concrete (Design Mix) in open foundations, with form work, at level. Cement concrete for plain/reinforced concrete in open foundations as per drawing and technical specification. [MORTH Specification: Section 1700 & 1500). With 20mm nominal size aggregate P.C.C. M-15 grade | | | | | | | | | | |
| | Abutment A1 | 1 | x | 8.20 | x | 8.50 | x | 0.75 | = | 52.28 | Cum |
| | Piers | 1 | x | 5.60 | x | 12.80 | x | 0.75 | = | 53.76 | Cum |
| | | 1 | x | 5.60 | x | 12.80 | x | 0.75 | = | 53.76 | Cum |
| | | 1 | x | 5.60 | x | 12.80 | x | 0.75 | = | 53.76 | Cum |
| | Abutment A2 | 1 | x | 8.20 | x | 8.50 | x | 0.85 | = | 59.25 | Cum |
| | | | | | | **Total Qty.** | | | | **272.80** | **Cum** |
| 8 | Structural cement concrete for plain concrete/ reinforced concrete for substructure at any level, complete as per [MORTH Specification: Section 1500 & 1700]. drawing and technical specification. Design mix concrete PCC M-25 Upto 5 mtr. Height | | | | | | | | | | |
| | Abutment A1 | 1 | x | 8.50 | x | 1.05 | x | 5.10 | = | 45.52 | Cum |
| | Return Wall | 2 | x | 5.50 | x | 0.75 | x | 6.50 | = | 53.63 | Cum |
| | Abut Cap | 1 | x | 8.80 | x | 0.90 | x | 0.40 | = | 3.17 | Cum |
| | Piers | 1 | x | 5.60 | x | 12.80 | x | 0.75 | = | 53.76 | Cum |
| | | 1 | x | 5.60 | x | 12.80 | x | 0.75 | = | 53.76 | Cum |
| | | 1 | x | 5.60 | x | 12.80 | x | 0.75 | = | 53.76 | Cum |
| | Abutment A2 | 1 | x | 8.50 | x | 1.05 | x | 5.10 | = | 45.52 | Cum |
| | Return Wall | 2 | x | 5.50 | x | 0.75 | x | 7.25 | = | 59.81 | Cum |
| | Abut Cap | 1 | x | 8.80 | x | 0.90 | x | 0.40 | = | 3.17 | Cum |
| | | | | | | **Total Qty.** | | | | **372.09** | **Cum** |
| 9 | Steel reinforcement in substructure HYSD reinforcement | | | | | | | | | | |
| | Abutment A1 & A2 | 2 | x | 3624 | | | | | = | 7248 | Kg |
| | Piers | 3 | x | 16637 | | | | | = | 49911 | Kg |
| | | | | | | **Total Qty.** | | | | **57159** | **Kg** |
| | | | | | | | | | | **57.16** | **MT** |
| 10 | Providing A.C. pipe Weep holes 150mm B-3 dia in Masonry/plain Concrete/Reinforced 13 Concrete abutment, wing wall/return wall complete as per drawing and technical specification. [MORTH Specification : Clause 2205]. | | | | | | | | | | |
| | Abutments | 2 | x | 7 | x | 5.25 | | | = | 73.50 | RM |
| | | 2 | x | 7 | x | 4.50 | | | = | 63.00 | RM |
| | | 2 | x | 7 | x | 3.50 | | | = | 49.00 | RM |
| | | 2 | x | 7 | x | 2.50 | | | = | 35.00 | RM |
| | | 2 | x | 7 | x | 1.50 | | | = | 21.00 | RM |
| | Return Walls | 4 | x | 5 | x | 3.50 | | | = | 70.00 | RM |
| | | 4 | x | 5 | x | 2.50 | | | = | 50.00 | RM |
| | | 4 | x | 5 | x | 1.50 | | | = | 30.00 | RM |
| | | 4 | x | 5 | x | 0.85 | | | = | 17.00 | RM |

| S.N. | Particulars | Nos. | | Length | | Breadth | | Height | | Qty. | Units |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | **Total Qty.** | | | | **408.50** | **RM** |
| 11 | Providing and and filling in foundation Btrenches, and back filling behind abutments, wing walls, return wall etc., and below pipe bed in layers not exceeding 150mm thick including watering and compacting (IRC-78) | | | | | | | | | | |
| | | 1 | x | 1.80 | x | 2.00 | x | 0.30 | = | 1.08 | Cum |
| | | 1 | x | 2.60 | x | 2.80 | x | 0.30 | = | 2.18 | Cum |
| | | 1 | x | 2.50 | x | 2.70 | x | 0.30 | = | 2.03 | Cum |
| | | 1 | x | 2.10 | x | 2.30 | x | 0.30 | = | 1.45 | Cum |
| | | 1 | x | 2.20 | x | 2.50 | x | 0.30 | = | 1.65 | Cum |
| | | 1 | x | 2.30 | x | 2.50 | x | 0.30 | = | 1.73 | Cum |
| | | | | | | **Total Qty.** | | | | **10.11** | **Cum** |
| 12 | Filter medium behind abutment, wing wall and return wall, complete as per drawing and technical specification. (Same as R7.20.3) [MORTH Specification: Clause 2504]. | | | | | | | | | | |
| | Abut | 2 | x | 7.90 | x | 3.10 | x | 4.50 | = | 220.41 | Cum |
| | Return | 4 | x | 4.80 | x | 2.30 | x | 4.50 | = | 198.72 | Cum |
| | | | | | | **Total Qty.** | | | | **419.13** | **Cum** |
| 13 | Structural RCC/PSC upto a height of 5 mtr. for reinforced concrete in superstructure complete including steel formwork, scaffolding etc. As per drawing and technical specification.Design mix concrete. [MORTH Specification : 1500 & 1700].RCC M-30 grade Solid slab superstructure at any level in superstructure complete as per drawing and technical Specification. [MORTH Specification: Section 1600] | | | | | | | | | | |
| | Slab | 4 | x | 10.00 | x | 8.50 | x | 0.77 | = | 261.80 | Cum |
| | Ends | 4 | x | 10.00 | x | 0.79 | x | 0.56 | = | 17.70 | Cum |
| | Kerb | 2 | x | 51.00 | x | 0.75 | x | 0.23 | = | 17.60 | Cum |
| | | | | | | **Total Qty.** | | | | **297.09** | **Cum** |
| 14 | Steel reinforcement in Superstructure HYSD reinforcement AS PER DETAILS ENCLOSED FOR 4 SLABS | | | **27064** | | **Kg** | | | | **27.06** | **MT** |
| 15 | Reinforced cement concrete wearing coat B-4 M-30 grade at any level including formwork and reinforcement complete as per drawing and technical Specification. MORTH Specification: Clause 2204, 1500 & 1700]. | | | | | | | | | | |
| | | 1 | x | 51.25 | x | 8.50 | x | 0.10 | = | 43.56 | Cum |
| | | | | | | **Total Qty.** | | | | **43.56** | **Cum** |
| 16 | Providing and Laying PCC M-15 grade B-4 leveling course below approach slab 15 complete as per drawing and technical specifications. [MORTH specification: 2700] | | | | | | | | | | |
| | | 2 | x | 7.90 | x | 4.50 | x | 0.20 | = | 14.22 | Cum |
| | | | | | | **Total Qty.** | | | | **14.22** | **Cum** |

| S.N. | Particulars | Nos. | | Length | | Breadth | | Height | | Qty. | Units |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 17 | Reinforced cement concrete M-30 grade approach slab including reinforcement 16 and formwork complete as per drawing and technical specifications. [MoRTH specification: 1500, 1600, 1700, 2704] | | | | | | | | | | |
| | | 2 | x | 7.90 | x | 4.50 | x | 0.30 | = | 21.33 | Cum |
| | | | | | **Total Qty.** | | | | | **21.33** | **Cum** |
| 18 | Providing, laying and fixing of Filler Joint Providing and fixing 2mm thick Corrugated Copper Plate in expansion joint complete as per drawing and technical specifications in MORTH specification: clause 2605. | | | | | | | | | | |
| | | 5 | x | 8.50 | x | 0.75 | | | = | 31.88 | Sq m |
| | | | | | **Total Qty.** | | | | | **31.88** | **Cum** |

| | | | |
|---|---|---|---|
| STRUCTUR SL | NO. OF UNITS | WIIGHT PER UNIT | TOTAL WEIGHT IN |
| PIER FOOTING | 3 | 14039 | 42117 |
| ABUTMENT FOOTING | 2 | 11215 | 22430 |
| PIER | 3 | 16637 | 49911 |
| ABUTMENT | 2 | 3624 | 7248 |
| DECK SLAB | 1 | 27064 | 27064 |
| | | TOTAL | 148770 |
| | | | |

| SCHEDULE OF REINFORCEMENT FOR ABUTMENT FOOTING | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| TYPE OF BAR | SHAPE | LOCATION | DIRECTION | DIA OF BAR IN MM | SPACING IN MM | NO. OF BARS | LENGTH OF EACH BAR (M) | TOTAL LENGTH (M) | Unit Weight Kg/M | TOTAL WEIGHT KgS. |
| J | | FOOTING BOTTOM | LONGITUDINAL | 25 | 125 | 68 | 8.90 | 605.2 | 3.86 | 2337 |
| K | | FOOTING BOTTOM | TRANSVERSE | 25 | 125 | 68 | 9.30 | 632.4 | 3.86 | 2442 |
| L | | FOOTING TOP | LONGITUDINAL | 20 | 125 | 68 | 8.90 | 605.2 | 2.47 | 1495 |
| M | | FOOTING TOP | TRANSVERSE | 20 | 125 | 68 | 9.30 | 632.4 | 2.47 | 1563 |
| M | | FOOTING STIRRUPS | TRANSVERSE | 16 | 150 | 570 | 3.75 | 2138 | 1.58 | 3378 |
| | | TOTAL IN ONE FOOTING | | | | | | | | 11215 |

| SCHEDULE OF REINFORCEMENT FOR PIER FOOTING | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| TYPE OF BAR | SHAPE | LOCATION | DIRECTION | DIA OF BAR IN MM | SPACING IN MM | NO. OF BARS | LENGTH OF EACH BAR (M) | TOTAL LENGTH (M) | Unit Weight Kg/M | TOTAL WEIGHT KgS. |
| J | | FOOTING BOTTOM | LONGITUDINAL | 25 | 125 | 45 | 14.40 | 648 | 3.86 | 2502 |
| K | | FOOTING BOTTOM | TRANSVERSE | 25 | 125 | 103 | 9.30 | 957.9 | 3.86 | 3698 |
| L | | FOOTING TOP | LONGITUDINAL | 20 | 125 | 45 | 14.40 | 648 | 2.47 | 1601 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| M | | FOOTING TOP | TRANSVERSE | 20 | 125 | 103 | 9.30 | 957.9 | 2.47 | 2367 |
| M | | FOOTING STIRRUPS | TRANSVERSE | 16 | 150 | 570 | 3.75 | 2138 | 1.58 | 3378 |
| | overlaps | | | 25 | | 45 | 2.08 | 93.6 | 3.86 | 362 |
| | overlaps | | | 20 | | 32 | 1.65 | 52.8 | 2.47 | 131 |
| | | TOTAL IN ONE FOOTING | | | | | | | | 14039 |
| | | | | | | | | | | |

**SCHEDULE OF REINFORCEMENT FOR PIER**

| TYPE OF BAR | SHAPE | LOCATION | DIRECTION | DIA OF BAR IN MM | SPACING IN MM | NO. OF BARS | LENGTH OF EACH BAR (M) | TOTAL LENGTH (M) | Unit Weight Kg/M | TOTAL WEIGHT KgS. |
|---|---|---|---|---|---|---|---|---|---|---|
| N | | PIER FLARED BOTTOM | VERTICAL | 25 | 125 | 136 | 7.51 | 1021 | 3.86 | 3941 |
| O | | PIER STRAIGHT | VERTICAL | 25 | 125 | 166 | 10.66 | 1770 | 3.86 | 6831 |
| P | | PIER | LATERAL | 20 | 125 | 45 | 31.17 | 1403 | 2.47 | 3465 |
| Q | | PIER FLARED BOTTOM | LATERAL | 20 | 125 | 5 | 37.34 | 186.7 | 2.47 | 462 |
| R | | PIER | CLOSED TIES | 16 | 150 | 540 | 1.63 | 877.5 | 1.58 | 1387 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| S | | PIER | OPEN TIES | 8 | EVERY 3 rd BAR BOTH WAYS | 1080 | 1.27 | 1376 | 0.40 | 551 |
| | | TOTAL IN ONE PIER | | | | | | | | 16637 |
| | | | | | | | | | | |
| **SCHEDULE OF REINFORCEMENT FOR ABUTMENT** | | | | | | | | | | |
| TYPE OF BAR | SHAPE | LOCATION | DIRECTION | DIA OF BAR IN MM | SPACING IN MM | NO. OF BARS | LENGTH OF EACH BAR (M) | TOTAL LENGTH (M) | Unit Weight Kg/M | TOTAL WEIGHT KgS. |
| N | ABUTMENT FACE | SKIN - VERTICAL | VERTICAL | 12 | 125 | 136 | 7.50 | 1020 | 0.89 | 908 |
| O | | SKIN - HORIZONTAL | VERTICAL | 12 | 125 | 166 | 8.50 | 1411 | 0.89 | 1256 |
| O | RETURN WALL FACE | SKIN - VERTICAL | VERTICAL | 12 | 125 | 160 | 5.00 | 800 | 0.89 | 712 |
| P | | SKIN - HORIZONTAL | LATERAL | 12 | 125 | 160 | 5.25 | 840 | 0.89 | 748 |
| | | TOTAL IN ONE ABUTMENT' | | | | | | | | 3624 |

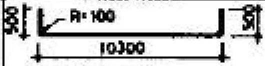
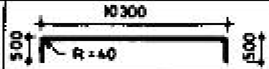
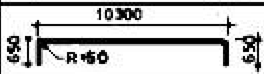
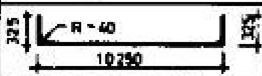
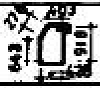
| SCHEDULE OF REINFORCEMENT FOR PIER CAP | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| TYPE OF BAR | SHAPE | LOCATION | DIRECTION | DIA OF BAR IN MM | SPACING IN MM | NO. OF BARS | LENGTH OF EACH BAR (M) | TOTAL LENGTH (M) | Unit Weight Kg/M | TOTAL WEIGHT KgS. |
| a | | VER STIRRUPS IN LONG DIRECTION | LONGITUDINAL | 20 | | 5 | 36.97 | 231.1 | 2.47 | 571 |
| s1 | | STIRRUPS IN SHORT DIRECTION | STIRRUPS | 12 | 300 | 61 | 5.35 | 407.9 | 0.89 | 364 |
| s2 | | PROJECTED PORTION | STIRRUPS | 12 | 100 | 30 | 4.75 | 178.1 | 0.89 | 159 |
| h1 | | HORIZONTAL STIRRUPS IN LONG DIRECTION | STIRRUPS | 12 | 240 | 3 | 38.75 | 145.3 | 0.89 | 130 |
| | | TOTAL IN ONE PIER CAP | | | | | | | | 1224 |

| SCHEDULE OF REINFORCEMENT FOR ABUTMENTR CAP | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| TYPE OF BAR | SHAPE | LOCATION | DIRECTION | DIA OF BAR IN MM | SPACING IN MM | NO. OF BARS | LENGTH OF EACH BAR (M) | TOTAL LENGTH (M) | Unit Weight Kg/M | TOTAL WEIGHT KgS. |
| a | | BOTTOM | LONGITUDINAL | 16 | | 8 | 18.40 | 184 | 1.58 | 291 |
| b | | TOP | LONGITUDINAL | 16 | | 8 | 18.40 | 184 | 1.58 | 291 |
| c | | | STIRRUPS | 12 | 300 | 61 | 2.55 | 194.4 | 0.89 | 174 |
| d | | SIDE FACE | | 12 | | 4 | 19.70 | 98.5 | 0.89 | 88 |
| | | TOTAL IN ONE ABUTMENT CAP | | | | | | | | 844 |

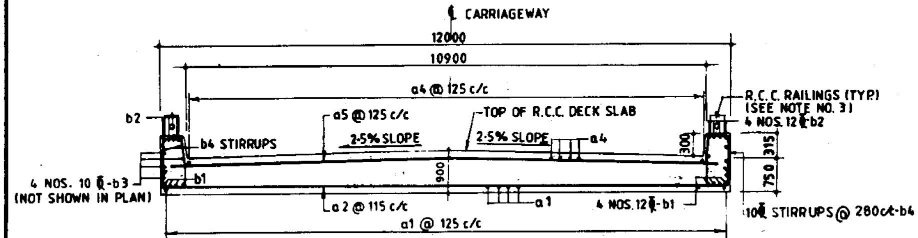
**SCHEDULE OF REINFORCEMENT (PER SPAN)** **DRG. NO.SD/114**

| SL.No. | LOCATION | BAR MKD | SHAPE | DIA (m m) | SPACING (m m) | LENGTH (m m) | NOS | TOTAL LENGTH(m) | WEIGHT (Kg.) |
|---|---|---|---|---|---|---|---|---|---|
| 1 | SLAB BOTTOM ALONG- X | a1 | | 25 | 125 | 11150 | 67 | 747.05 | 2914 |
| 2 | SLAB BOTTOM ALONG- Y | a2 | | 12 | 115 | 8830 | 94 | 830.02 | 748 |
| 3 | SLAB SIDE ALONG-Y | a3 | | 10 | | 11900 | 2 | 23.80 | 17 |
| 4 | SLAB TOP ALONG- X | a4 | | 10 | 125 | 11240 | 64 | 719.36 | 504 |
| 5 | SLAB TOP ALONG- Y | a5 | | 10 | 125 | 9000 | 84 | 756.00 | 530 |
| 10 | KERB- BOTOM | b1 | | 12 | - | 11500 | 8 | 92.00 | 83 |
| 11 | KERB- TOP | b2 | | 12 | - | 11530 | 8 | 92.24 | 84 |
| 11 | KERB- SIDES | b3 | | 10 | - | 10840 | 8 | 86.72 | 61 |
| 12 | KERB- STIRRUPS | b4 | | 10 | 200 | 2870 | 76 | 218.12 | 153 |
| 13 | Wearing Course X | | | 12 | 150 | 7900 | 80 | 632.00 | 569 |
| 14 | Wearing Course y | | | 12 | 150 | 10000 | 67 | 670.00 | 603 |
| | | | | | | | | | 6266 |
| | | | **qty for 4 slabs** | | | **4** | **x** | **6266.00** | **25064** |
| | OVERLAPS | | | | | | | | |
| | SLAB BOTTOM ALONG- X | a1 | SLAB 1,2,3 AND 4 (28+26+30+32) | 25 | | 2050 | 116 | 237.80 | 928 |
| | SLAB BOTTOM ALONG- Y | a2 | SLAB 1,2,3 AND 4 (31+33+27+33) | 12 | | 9600 | 124 | 1190.40 | 1072 |
| | | | | | | | TOTAL | | **27064** |

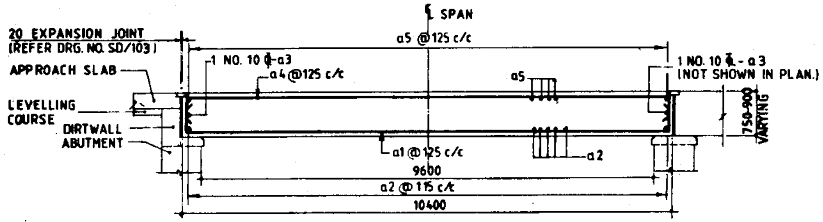
# SCHEDULE OF REINFORCEMENT (PER SPAN)

| SL.NO | LOCATION | BAR MKD | SHAPE | DIA. (mm) | SPACING (mm) | LENGTH (mm) | NOS. | TOTAL LENGTH(m) | WEIGHT (kg) |
|---|---|---|---|---|---|---|---|---|---|
| 1 | SLAB-BOTTOM ALONG-X | a1 |  | 25 | 125 | 11150 | 96 | 1070.40 | 4121 |
| 2 | SLAB-BOTTOM ALONG-Y | a2 |  | 12 | 115 | 12430 | 91 | 1131.13 | 1004 |
| 3 | SLAB-SIDE ALONG-Y | a3 |  | 10 | — | 11900 | 2 | 23.80 | 15 |
| 4 | SLAB-TOP ALONG-X | a4 |  | 10 | 125 | 11240 | 89 | 1000.36 | 617 |
| 5 | SLAB-TOP ALONG-Y | a5 |  | 10 | 125 | 11600 | 84 | 974.40 | 601 |
| 6 | KERB-BOTTOM | b1 |  | 12 | – | 11500 | 8 | 92.00 | 82 |
| 7 | KERB-TOP | b2 |  | 12 | – | 11530 | 8 | 92.24 | 82 |
| 8 | KERB-SIDES | b3 |  | 10 | – | 10840 | 8 | 86.72 | 54 |
| 9 | KERB-STIRRUPS | b4 |  | 10 | 200® | 2870 | 76® | 218.12 | 135 |

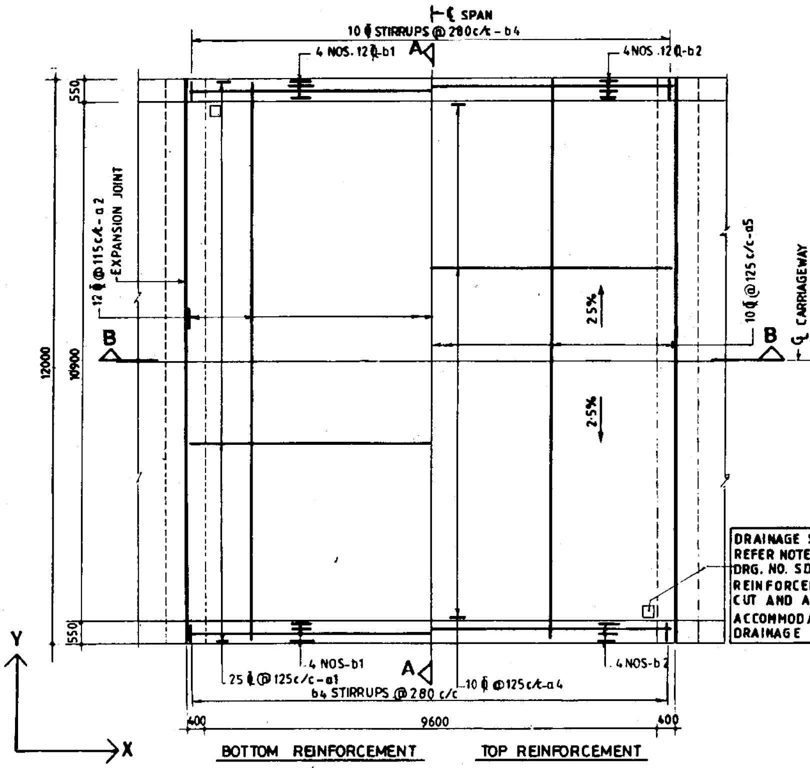
® DOES NOT INCLUDE THE ADDITIONAL STIRRUPS AT LAPS OF LONGITUDINAL BARS-REFER DRG NO SD/103




## SECTION A-A




## SECTION B-B *




DRAINAGE SPOUT(TYP.) REFER NOTE NO. 2 OF DRG. NO. SD/102. REINFORCEMENT TO BE CUT AND ADJUSTED TO ACCOMMODATE THE DRAINAGE SPOUT.

## PLAN SHOWING REINFORCEMENT DETAILS OF DECK SLAB *

## SCHEDULE OF REINFORCEMENT (PER SPAN)

| SL.NO | LOCATION | BAR MKD | SHAPE | DIA. (mm) | SPACING (mm) | LENGTH (mm) | NOS. | TOTAL LENGTH(m) | WEIGHT (kg) |
|---|---|---|---|---|---|---|---|---|---|
| 1 | SLAB-BOTTOM ALONG-X | a1 | 500, R=100, 10300, 500 | 25 | 125 | 11150 | 96 | 1070.40 | 4121 |
| 2 | SLAB-BOTTOM ALONG-Y | a2 | 300, R=80, 11900, 300 | 12 | 115 | 12430 | 91 | 1131.13 | 1004 |
| 3 | SLAB-SIDE ALONG-Y | a3 | 11900 | 10 | — | 11900 | 2 | 23.80 | 15 |
| 4 | SLAB-TOP ALONG-X | a4 | 10300, 500, R=40, 500 | 10 | 125 | 11240 | 89 | 1000.36 | 617 |
| 5 | SLAB-TOP ALONG-Y | a5 | 5800, 5800, 145 | 10 | 125 | 11600 | 84 | 974.40 | 601 |
| 6 | KERB-BOTTOM | b1 | 650, R=50, 10275, 650 | 12 | — | 11500 | 8 | 92.00 | 82 |
| 7 | KERB-TOP | b2 | 10300, 650, R=50, 650 | 12 | — | 11530 | 8 | 92.24 | 82 |
| 8 | KERB-SIDES | b3 | 325, R=40, 10250, 325 | 10 | — | 10840 | 8 | 86.72 | 54 |
| 9 | KERB-STIRRUPS | b4 | 643, 916, 240, 403, 426 | 10 | 280® | 2870 | 76® | 218.12 | 135 |

® DOES NOT INCLUDE THE ADDITIONAL STIRRUPS AT LAPS OF LONGITUDINAL BARS-REFER DRG. NO. SD/103

Dead load of the superstructure per span including R.C.C. railings @ 3 kN/m and wearing coat @ 2 kN/sq m = 2843 kN.

## REFERENCE DRAWINGS

| DRAWING NO. | TITLE |
|---|---|
| SD/101 | GENERAL NOTES |
| SD/102 | GENERAL ARRANGEMENT |
| SD/103 & SD/104 | MISCELLANEOUS DETAILS |
| SD/105 | DETAILS OF R.C.C. RAILINGS. (WITHOUT FOOTPATHS) |

## QUANTITIES (PER SPAN)

| | |
|---|---|
| EFFECTIVE SPAN (m) | 10.0 |
| CONCRETE (cu. m.) | 106.40 |
| STEEL INCLUDING 5% EXTRA FOR LAPS AND WASTAGE (kg) | 7047 |
| ASPHALTIC WEARING COAT (sq. m) | 113.36 |

## NOTES:-

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. Special attention is invited to note, no.(B)4 of drawing no SD/101 regarding the design mix to be adopted.
3. The reinforcements of railing posts shall be incorporated before casting of the deck slab. The railings shall conform to drawing no SD/105 or any other approved type.
4. Dimensions in schedule of reinforcement are given as per IS 2502
* 5. Reinforcement of adjacent span superstructure, approach slab not shown.

| MKD. | DATE | DESCRIPTION | BY |
|---|---|---|---|
| | | | |
| | | | |

REVISION

GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

STANDARD DRAWINGS FOR ROAD BRIDGES

R.C.C. SOLID SLAB
SUPERSTRUCTURE (RIGHT) SPAN 10·0m
(WITHOUT FOOTPATHS)

| RECOMMENDED BY | | APPROVED BY | 1990 |
|---|---|---|---|
| Reddy (S.S. REDDY) E.E. | (S.K. KAISTHA) S.E. | (P.N. SHIVAPRASAD) C.E. | DRG NO SD/114 |